SOP Number and version: SOP-NAT-STER.v01

SOP Title: **Sterilization & Disinfection**

Table 1. SOP current version review and approval table

	NAME	TITLE	SIGNATURE	DATE
Author	Research name	Student		01/10/2024
Reviewer	Baris Ata Borsa	Lab responsible		25/11/2024
Approval	Frank J. Hernandez	Principal Investigator		09/01/2025

Effective Date:	**09/01/2025**
Review Date	09/01/2025

Approved by PI (Frank J. Hernandez) - 2026-02-17T12:29:59.286Z

TABLE OF CONTENTS

TABLE OF CONTENTS

1. Overview 2
2. Scope.....2
3. Procedure.....2
4. Purpose/Objective 7
5. Definition/Explanation of key terms 7
6. Responsibilities/Process owner.....8
7. Equipment/Material 8
8. Caution/Warning.....8
9. References 8
10. Version control 8
11. Attachments/Appendices 9

1. Overview and background

Thorough application of sterilization and disinfection practices is crucial for lab safety as well as accuracy of the experiment results. In a BSL-2 laboratory, sterilization and disinfection practices must be well characterized and documented.

2. Scope

In this SOP, sterilization and disinfection practices used in Nucleic Acids Technology laboratories located at Linköping University (Campus Valla, B-huset, Rum 2E:686) are explained.

This SOP covers the sections below.

- Autoclave sterilization
- Surface disinfection
- Hand disinfection
- Waste disposal

3. Procedure

3.1. Autoclave sterilization

Autoclave is an instrument used to sterilize material by using high temperature and pressure. Nucleic Acids Technology laboratories staff use the autoclave in the Campus Valla, B-huset, Room 224:193 for this purpose.

- Turn on the main switch to power up the machine.
- Place the material to be autoclaved in one of the devoted baskets and place it carefully inside the autoclave.
- If the material is solid, the thermometer should be left in its holder on the inside of the autoclave lid.
- If the material is liquid, a temperature control bottle (containing water) should be placed inside the basket with the rest of the waste and the thermometer should be introduce inside the liquid, making sure that the metal tip is fully submerged in water.
- Fill the glassware only half full when sterilizing liquids.
- To prevent bottles from shattering during pressurization, the caps of vessels with liquids must be fully loosened before loading.
- Close the lid by applying pressure on it and wait for the system to self-lock the lid.
- Navigate through the menu system using the control panel buttons and select the program that corresponds to your material type (e.g., liquid, solid, waste) and set the program.
- Press start and make sure that the cycle has started correctly. Check the information note printed by the autoclave.
- Once the cycle is finished, press open in the control panel. The lid will open automatically.
- Open the lid fully and let the content cool down for 10 minutes.
- Always use protective gloves to remove the contents from the autoclave.
- Remove the thermometer probe, the control bottle and take the basket out of the autoclave before removing the sterile content.
- Clean the basket and return it to its corresponding shelf.
- Report the use in the logbook that is placed on the shelf next to the autoclave.
- Rip and store the printed report in the dedicated folder that is placed on the shelf next to the autoclave.
- Always use autoclaving tape as a control for the sterilization of the products and as a label for sterile products.

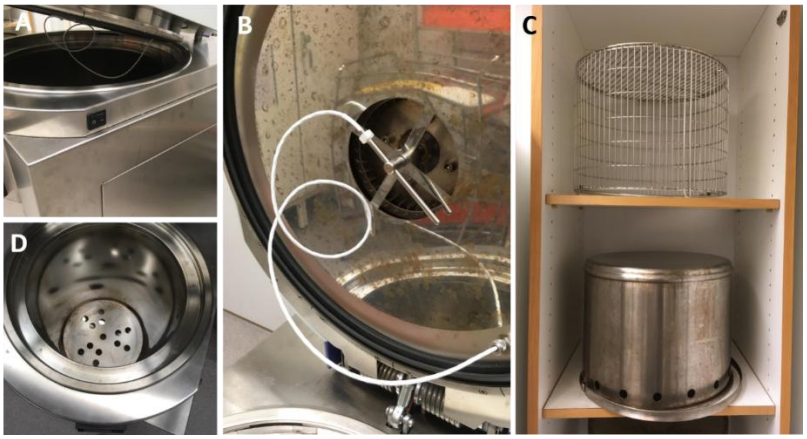

Figure 1: A) Main switch. B) Thermometer on lid holder. C) Baskets for new autoclave. D) Autoclave chamber.

3.2. Surface disinfection

Alcohol based disinfectants are used for surface disinfection. Ethanol (ethyl alcohol) and 2-propanol (isopropyl alcohol) have similar disinfectant properties. They are active against vegetative bacteria, fungi, and lipid-containing viruses but not against spores. Their action on non-lipid-containing viruses is variable. For highest effectiveness they should be used at concentrations of approximately 70% in water.

3.2.1. Regular surface disinfection before and after use

- Spray alcohol-based disinfectant to the lab surface to be disinfected
- Wait for 10 minutes
- Wipe with a clean paper towel
- Discard the paper towel in biological waste bag

3.2.2. Surface disinfection after contamination with infectious material

- Carefully remove the visible contaminant material (e.g., spilled liquid, tissue etc.) using a paper towel
- Discard the paper towel in biological waste bag
- Cover the entire area of contamination by pouring alcohol-based disinfectant
- Wait for 10 minutes
- Wipe with a clean paper towel
- Discard the paper towel in biological waste bag

3.3. Hand disinfection

3.3.1. Hand washing with soap and water

Wash your hands with soap and water before entering and leaving the lab as well as when they are visibly soiled/dirty.

- Wet hands with water
- Apply the soap to your hands
- Rub hands together vigorously for at least 15 seconds, covering all surfaces of the hands and fingers
- Rinse hands with water and use disposable towels to dry. Use a towel to turn off the faucet
- Avoid using hot water to prevent drying of the skin

3.3.2. Hand disinfection with alcohol-based hand disinfectant

Use alcohol-based hand disinfectant when you cannot wash your hands, after a clear contamination with an infectious material, and before leaving the lab (after hand washing).

- Put product on hands and rub hands together.
- The efficacy of alcohol-based hand disinfectant depends on the volume applied to the hands. Use the recommended amount of alcohol-based hand disinfectant product to clean your hands.
- Cover all surfaces and rub until hands feel dry.
- This should take around 20 seconds.
- Pay attention to the areas frequently missed (e.g., thumbs, fingertips, between fingers)

Figure 2. Steps of accurate hand disinfection.

3.4. Waste disposal

Always separate liquid waste from solid waste.

3.4.1. Liquid waste disposal

- Inactivation of liquid biological waste can be achieved by autoclaving (121 ^{o}C for 15-20 minutes or 115 ^{o}C for 30 minutes) or by treating the solutions with an appropriate chemical decontaminant (e.g. 70% ethanol, <5% sodium hypochlorite).
- Decontaminated liquid waste can be poured into sink unless it contains other harmful chemicals (e.g., antibiotics). If it does, contact the Lab-safety department of the facility via labsafety@liu.se

3.4.2. Solid waste

- Solid waste must be separated as regular waste, biological waste and sharp waste
- Regular waste will be collected by the university staff
- All biological waste must be disposed in a double-layer biohazard waste bag for biological material **(Figure 3A)**
- When double-layer biohazard waste bag is full (do not use it for more than 70% of the volume), autoclave the bag to decontaminate the waste.
- After autoclaving, close the inner plastic bag properly with tape, then close the outside paper bag with seal as showed in the **Figure 3B**. Make sure the bag is properly closed.
- Take the double-layer biohazard waste bag to the waste collection room (Källsorteringsrum in Swedish). Room 1B:632 at the B building (Biology), first floor. The container tagged 56 is the one for materials that will be incinerated **(Figure 5).**
- Sharp waste will be disposed in the special single use puncture safe containers labeled with UN 3291 **(Figure 6).**
- Puncture safe containers will be placed on the lab bench and inside the biosafety cabinet.
- Sharps containers should not be filled above the indicated line, usually only the ¾ should be full.
- When the puncture safe container is full the container will be closed and left in the recycling room (Miljörum in Swedish)
- Order a pickup by using the webform

Figure 3. A) Double-layer biohazard waste bag. B) How to close a double-layer biohazard waste bag

Figure 4. How to dispose biological waste.

Figure 5. Sharps container

4. Purpose/Objective

The objective of this SOP is to guide Nucleic Acids Technology laboratories staff on sterilization and disinfection practices in Nucleic Acids Technology laboratories.

5. Definition / Explanation of key terms

- SOP – Standard Operating Procedure
- R&D – Research and Development
- CTO – Chief Technology Officer
- CSO – Chief Scientific Officer
- COO – Chief Operations Officer
- PPE – Personal Protective Equipment
- UN – United Nations

6. Responsibilities

This SOP was prepared by Student name, revised by Baris Ata Borsa and approved by Frank J. Hernandez, principal investigator.

7. Equipment/Material

- Autoclave
- Autoclave gloves
- Double-layer biohazard waste bags
- Sharps container
- Hand disinfectants
- Surface disinfectants
- Gloves
- Paper towel
- Water
- Soap

8. Caution/Warning

- Use PPE according to **SOP-NAT-BSL2**.

9. References

- https://www.cdc.gov/infection-control/hcp/disinfection-sterilization/steam-sterilization.html

- World Health Organization. Infection prevention and control of epidemic- and pandemic-prone acute respiratory infections in health care. (2014). ISBN 978 92 4 150713 4.

- https://www.cdc.gov/clean-hands/about/index.html

- https://liu.se/en/lab-safety/waste/laboratory-waste

10. Version Control

Original version of the **SOP-NAT-STER.v01**. Check **Table 1**.

11. Attachments/Appendices

Appendix I:

- Autoclave manual